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Remote sensing of snow and ice

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Abstract. Monitoring of snow and ice on the Earth's surface will require increasing use of satellite remote sensing techniques. These techniques are evolving rapidly. Active and passive sensors operating in the visible, near infrared, thermal infrared, and microwave wavelengths are described in regard to general applications and in regard to specific USA or USSR satellites. Meteorological satellites (frequent images of relatively crude resolution) and Earth resources satellites such as Landsat (less frequent images of higher resolution) have been used to monitor the areal extent of seasonal snow, but problems exist with cloud cover or dense forest canopies. Snow mass (water equivalent) can be measured from a low-flying aircraft using natural radioactivity, but cannot yet be measured from satellite altitudes. A combination of active and passive microwave sensors may permit this kind of measurement, but not until more is known about radiation scattering in snow. Satellite observations are very useful in glacier inventories, correcting maps of glacier extent, estimating certain mass balance parameters, and monitoring calving or surging glaciers. Ground ice is virtually impossible to monitor from satellites; ice on rivers and lakes can be monitored only with very high-resolution sensors. Microwave sensors, due to their all-weather capability (the ability to see through clouds) provide exciting data on sea ice distribution. Analysis of digital tapes of satellite data requires the archiving and scanning of huge amounts of data. Simple methods for extracting quantitative data from satellite images are described.

La télédétection de neige et de glace

Résumé. La surveillance de neige et de glace sur la surface de la Terre nécessitera l'utilisation croissante des techniques de télédétection par satellites. Ces techniques s'élaborent de plus en plus vite. On décrit les détecteurs actifs et passifs opérant dans les fréquences visibles, presque infrarouges, infrarouges thermiques et microondes par rapport à des applications générales et par rapport à de spécifiques satellites des Etats Unis ou de l'URSS. Des satellites météorologiques (des images fréquentes de résolution assez crue) et des satellites qui mettent en surveillance les ressources terrestrielles tels que Landsat (des images moins fréquentes de résolution plus nette) ont été utilisés pour surveiller l'étendue de la zone de neige saisonnière, mais il y a encore des problèmes quand il s'agit de la converture des nuages ou des dômes des forêts épaisses. On peut mesurer la masse de neige (l'équivalent en eau) d'un avion qui vole à basse altitude en utilisant la radioactivité naturelle, mais jusqu'ici on ne peut pas le mesurer des altitudes où se trouvent les satellites. Une combinaison des détecteurs microondes passifs et actifs permettra peut-être ce genre de mesure à l'avenir, mais seulement lorsqu'on comprend mieux la diffusion par rayonnement relative à la neige. Les observations par satellite sont très utiles pour les inventaires des glaciers, pour corriger des cartes d'étendue des glaciers, pour l'estimation de certains paramètres de bilan de masse et pour mettre en surveillance des glaciers en velage ou décolmatage. Il est presque impossible de mettre en surveillance par des satellites la glace de fond; les détecteurs à très haute résolution seulement peuvent mettre en surveillance la glace sur les rivières et sur les lacs. Les détecteurs microondes, à cause de leurs capabilités par tous les temps (la capacité de pouvoir voir à travers les nuages) fournissent des données bien intéressantes sur la répartition de la glace de mer. L'analyse des bandes numériques des données des satellites nécessite l'archivage et la scrutation de vastes quantités de données. On décrit quelques méthodes simples pour l'extraction des données quantitatives des images des satellites.

INTRODUCTION

Snow and ice on the Earth's surface affect man in many direct and indirect ways. However, monitoring of snow and ice is difficult because snow and ice cover vast regions, including inaccessible areas, and may change very rapidly with time. The powerful new techniques of remote sensing are beginning to remove some of this difficulty, thereby opening up opportunities to make synoptic measurements of snow and ice and to utilize these data in many new and useful ways. Some examples of these new opportunities are:

(1) Forecasts of snowmelt runoff may be improved and thus reservoirs operated more efficiently.
(2) Soil and irrigation water supplies may be predicted with greater accuracy, allowing more productive planting and cropping schedules.
(3) Snowmelt flood potential may be anticipated more exactly and subsequent damage alleviated.
(4) Freezing and breakup of rivers and lakes may be monitored over large areas, permitting better control of river navigation and reservoir operation to minimize losses.
(5) Sea ice formation, movement, and breakup may be watched to optimize vessel traffic schedules and to carry out improved marine geophysical surveys for oil and minerals.
(6) Changes in climate may be detected over regional or global areas, and these data used for many scientific and economic purposes.
(7) Data on seasonal snow and sea ice cover may be used in general circulation and other climatic models to predict climatic variations.

Few of these important monitoring tasks are feasible today, at least in an operational (practical) sense. The results of research suggest that none of them is impossible and, in many cases, the transfer of technology from research to operation is well underway. Some of the listed tasks cannot be made possible until a new generation of satellites—particularly satellites employing all-weather microwave sensors—are in orbit. Some tasks may be performed with simple graphical or image-analysis techniques using inexpensive readout stations, but others may require powerful computers and expensive digital processing techniques. Nevertheless, the possibilities for the remote monitoring of snow and ice are exciting.

The International Commission on Snow and Ice (ICSI) of IAHS has long been interested in the use of remote sensing as a tool to accomplish its objectives. ICSI established three major new programmes during the International Hydrological Decade (IHD), all of which have been continued during the International Hydrological Programme (IHP). These programmes are: (1) World Inventory of Snow and Ice Masses, (2) Glacier Fluctuations, and (3) Combined Heat, Ice, and Water Balances at Representative Glacier Basins. All three programmes require the valuable tool of remote sensing. Understanding the complex links between ice and climate, the major theme of ICSI during the 1976–1979 quadrennium, requires even greater application of remote sensing techniques.

This survey presents a general description of the types of sensors and satellites available during the period 1976–1979, a discussion of how different types of snow and ice can or might be monitored, and some examples of the kinds of data reduction methodologies which can be employed. Attention is given to the use of satellite remote sensing, as satellite data cover the most area and are often least expensive to the user. Ice and snow on land is stressed because of its application to the IHP. The remote sensing of snow is further emphasized because (1) it is of great climatological significance; (2) it is by far of greatest practical importance in hydrology; and (3) it normally overlies other glaciological materials much of the year, so the problems of measuring snow are common to many other glaciological measurement programmes.

SENSORS AND SATELLITES

Remote sensing refers to the gathering of information about an object without physical contact with that object. Remote sensing applied to ice and snow refers in this paper to the gathering of information about the glaciological object (such as a snow pack) using a sensing device carried in an aircraft or satellite. The pertinent information is conveyed through the atmosphere by electromagnetic waves (radiation). The Earth's atmosphere is nearly opaque to radiation of some wavelengths, thereby eliminating such wavelengths from consideration or else restricting their application to measurements from very low-flying aircraft. At other wavelengths, however, the atmosphere is nearly transparent, allowing the use of sensors mounted in satellites. The radiant energy may originate in the object as natural radioactivity or thermal radiation; it may originate in the sun and be modified as it reflects off the object; or it may originate in the aircraft or satellite. The latter case, in which the object is illuminated by the sensing device, is called an active system; the first two are passive systems. Table 1 lists some of the more important sensing systems, by wavelength of radiation employed and by their active or passive nature.

Some of these sensors, such as gamma-ray scintillometers, can only be operated from low-flying aircraft and thus involve considerable expense for repetitive surveys. The use of sensors in satellites flown for meteorological, Earth resources, or oceanographic purposes is far more efficient and cost-effective for repetitive surveys of large areas than aircraft. Table 2 presents current information on USSR and USA civilian scientific (non-military or intelligence) sensors and their satellite platforms.

Satellite data can be obtained from the countries which operate the particular satellite, or from countries which operate ground-based readout facilities. For instance, Landsat data can be obtained from the EROS Data Center, US Geological Survey, Sioux Falls, South Dakota 57198, USA; readout stations are currently processing Landsat data in Brazil, Canada, and Italy, and other stations in Argentina, Australia, India, Iran, Japan, and Sweden are expected to begin processing data during the period 1978–1980. American meteorological satellite data are available through the National Environmental Satellite Service, World Weather Building, Camp Spring, Maryland 20031, USA.

TYPES OF SNOW AND ICE COVER

Seasonal snow cover

Seasonal snow is widespread, constantly changing in structure and in properties,

TABLE 1. Some remote sensing systems useful for snow and ice studies

Wavelength region	Typical sensing device	Spatial resolution* Existing	Planned	Application
1. Passive sensors				
Gamma rays	Scintillometer	—	—	Measures snow and ice mass directly. Requires very low flight altitudes (~100 m above terrain).
Visible	Film camera, television camera, or photomultiplier	30	15	Measures surface reflectance, distinguishes snow from ice and snow/ice from other materials. Several wavelengths may permit classifying snow surface character (wet/dry, etc.).
Near infrared†	Camera or photomultiplier	80	30	Measures surface reflectance, distinguishes snow from ice and snow/ice from other materials. Several wavelengths may permit classifying snow surface character (wet/dry, etc.). Also, may differentiate snow from cloud if wavelength 1.5–3 μm is used.
Thermal infrared	Radiometer	240	120	Primarily measures surface temperature. May be used to infer lake or sea ice thickness.
Microwave	Radiometer	30 000	20 000	Separates snow and ice from water and possibly from land, multi-year sea ice from new sea ice. Several wavelengths may distinguish snow wetness, may possibly permit measurement of grain size, density, mass, and layering.
2. Active sensors				
Visible	Laser altimeter	—	—	Measures surface roughness, altitude.
Microwave	Radar	25	25	Measures surface altitude, form, roughness, may possibly measure subsurface properties if several wavelengths used.
Microwave	Altimeter	1 600+	—	Measures variations in altitude of surface to within ±10 cm.

* Approximate values in metres, from normal satellite altitudes, and represent highest resolution of existing or planned sensor as of 1 July 1978, for unmanned spacecraft.
† Also called far red or photographic infrared.

irregular in thickness and in distribution, and rapid in formation and in appearance (Fig. 1). It also overlies most other glaciological materials much of the year. Because of its variability, snow cannot be monitored successfully, except in very limited situations, without the use of remote sensing technology.

The areal extent of a snow cover is an important property to monitor (UNESCO/IAHS/WMO, 1970). Measurement of this property, using aircraft or satellite remote sensing devices, is relatively straightforward because snow is so reflective

TABLE 2. USSR and USA civilian scientific satellites and sensors useful for snow and ice studies (from WMO, 1976, with minor corrections)

A. USSR

METEOR satellites are equipped with the following sensors providing operational information:

(1) Television cameras, sensitive to radiation in the visible spectral band, for obtaining data relating to the amount of cloud, ice, and snow cover. Resolution at perigee is 1.5 km and the swath covered along the orbit is about 1400 km wide;

(2) Infrared television-type instrumentation for obtaining data relating to amount of cloud and ice cover. Resolution at perigee is 20 km, the swath covered along the orbit is about 1600 km wide and the spectral range is 8–12 μm;

(3) Scanning radiometer for measuring the intensity of reflected radiation (spectral range 0.3–3 μm) and outgoing radiation (spectral range 3–30 μm) and the temperature of the underlying surface and of cloud tops (spectral range 8–12 μm). Resolution at perigee is 70 km and the swath covered along the orbit is 2700 km wide.

In addition, the experimental series of METEOR satellites are equipped with the following sensors:

(1) Scanning telephotometer for automatic direct transmission of pictures in the visible spectral band. The resolution at perigee is 1.6 km, and the swath covered along the orbit is 2700 km wide;

(2) Eight-channel scanning radiometer for measuring vertical temperature profiles. The resolution at perigee is 13×53 km, and the spectral (wavelength) channels are 11.1, 13.33, 13.70, 14.24, 14.43, 14.75, 15.02, and 18.70 μm.

(3) Passive microwave instrumentation for detecting areas of precipitation and ice cover. The resolution at perigee is 50 km, and the wavelength is 0.8 cm.

(4) Four-channel television-type scanning instrumentation for obtaining multispectral images of cloudiness and the underlying surface, the spectral channels are 0.5–0.6, 0.6–0.7, 0.7–0.8, and 0.8–1.1 μm.

B. USA (updated to 1 July 1978)

Sensor	Spacecraft	Nominal spectral band (μm unless specified)	Nominal resolution at nadir (km)	Swath width (km)	Repeat coverage	Remarks
SR (Scanning radiometer)	NOAA-5, 6, . . . (operational)	0.40–1.1 10.5–12.5	4.2 8	3400	12 h	Global coverage.
VHRR (Very high resolution radiometer)	NOAA-5, 6, . . . (operational)	0.6–0.7 10.5–12.5	1	3400	12 h	Mostly direct readout; limited stored data capability that can be assigned to polar regions.

AVHRR (Advanced VHRR)	NOAA (TIROS-N) (Operational first launch in 1978)	0.55–0.9 0.225–1.0 3.55–3.93 10.5–11.5	1 4	2800	12 h	Global coverage at lower resolution. Mostly direct readout at higher resolution; more stored data in polar regions than VHRR. Platform location and data relay system.
VISSR (Visible/infrared spin-scan radiometer)	Goes-3, 4, . . . (operational)	0.55–0.70 10.5–12.6	0.80 8	3500 (at lat. 50°)	$\frac{1}{2}$ h	Polar coverage limited to about 60° latitude because of equatorial orbit. Spatial resolutions more than double nadir value 50° lat. Data relay system. Temperature range 180–315 K.
ESMR (Electrically-scanning microwave radiometer)	Nimbus-5, 6	1.55 cm	32	1100	12 h	Near global coverage. Platform location and data relay system.
THIR (Temperature-humidity infrared radiometer)	Nimbus-6 Nimbus-G (1978 launch)	6.5–7.1 10.5–12.5	22 0.8	1800	12 h	Global coverage.
SMMR (Scanning multi-frequency microwave radiometer)	Nimbus-G Seasat-1	0.8–6 cm (5 bands, 10 channels)	16–87 as a function of frequency	78 (Nimbus-G) 900 (Seasat-1)	36 h	Global coverage for Nimbus-G. Polar coverage for Seasat-1 limited to latitudes less than about 72°. Only 0.8 cm channel expected to yield ice information.
SAR (Synthetic aperture radar)	Seasat-1	22 cm	0.25	100	36 h	Direct readout only; limited swath lengths.
VIR (Visible/infrared radiometer)	Seasat-1	0.49–0.94 10.5–12.5	3 5	679	36 h	Polar coverage restricted (see SMMR).
MSS (Multi-spectral scanner system)	Landsat-1, 2, 3, D	0.5–0.6 0.6–0.7 0.7–0.8	0.8	185	18 days	Polar coverage limited to latitudes less than about 82°; data swaths overlap for several

		0.8–1.1 4–12.6 (Landsat-3)	0.240			successive days in each cycle at latitudes greater than about 70°. Data relay system. Two satellites presently provide coverage twice during an 18-day period.
RBV (Return beam vidicon)	Landsat-1, 2	0.475–0.575 0.580–0.680 0.690–0.830	0.08	185	18 days	Polar coverage restricted (see MSS).
	Landsat-3	0.505–0.750	0.030	38 × 98		
HCMR (Heat capacity mapping radiometer)	HCMM-1	0.5–1.1 10.5–12.5	0.6	700	12 h	Direct readout only. Only polar coverage will be Beaufort, Chukchi, and Bering Seas south of 73° N. Temperature range 260–340 K.
CZCS (Coastal zone color scanner)	Nimbus-G	0.43–0.45 0.51–0.53 0.54–0.56 0.66–0.68 0.70–0.80 10.5–12.5	0.8	1800	4 days	Mostly direct readout.
Microwave scatterometer	Seasat-1	2.2 cm	50 (high gain) 200 (low gain)	280–750	36 h	Very few observations in 460-km wide band centered on nadir. Polar coverage restricted (see SMMR).
Radar altimeter	Seasat-1	2.2 cm	1.6–12	10	6 months	Polar coverage restricted (see SMMR).
Thematic mapper	Landsat-D (Follow on) (early 1981 launch via space shuttle)	0.45–0.52 0.52–0.60 0.63–0.69 0.76–0.90 1.55–1.75 2.08–2.35 10.4–12.5	0.030 0.030 0.030 0.030 0.030 0.030 0.120	185	18 days	Proposed 7–9 day coverage with two operating satellites. Data delivery in 48–96 h using all digital processing and communication satellites for global relay.

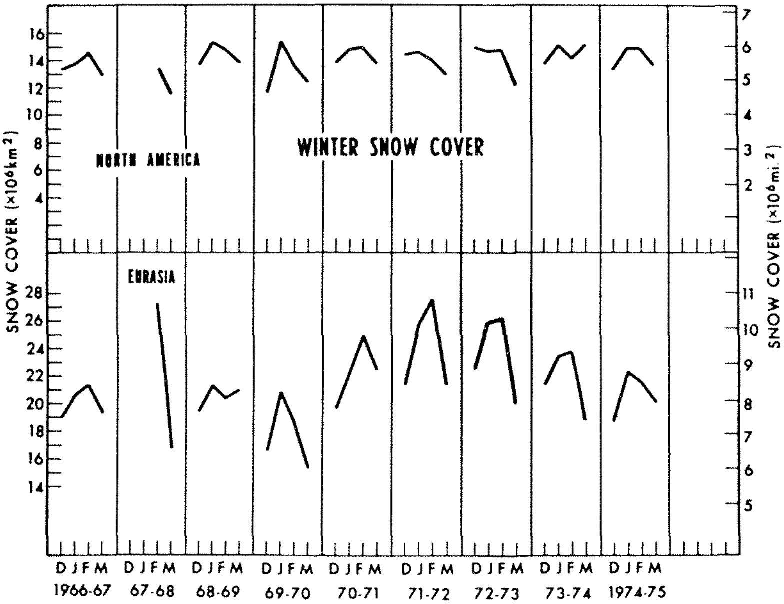

FIG. 1. Variations in northern hemisphere winter season snow cover, 1966–1975, as determined by satellite images (from Wiesnet & Mattson, 1976),

(white) compared with other materials on Earth (Figs 2 and 3). Sensors operating in the visible wavelength region have been used for many years to measure the areal extent of snow cover either from aerial photographs or from satellite images (Fritz, 1962; Barnes & Bowley, 1968; Wiesnet & Mattson, 1976). Meteorological satellites, such as the METEOR or NOAA series, pass over given areas on Earth at least once a day so that the rate of image acquisition is high. Synchronous meteorological satellites provide even more frequent coverage because they continuously scan one hemisphere of the Earth. However, none of these satellites has sufficient resolution to identify small snow patches or snow in forested areas, nor do they present images which can be directly related to conventional map projections. A serious deficiency of the synchronous satellites for monitoring snow cover is that they do not show much usable data beyond about 50° N or S latitude because the viewing direction is too oblique at high latitudes.

The Earth Resources Technology Satellite, ERTS (renamed Landsat by NASA in January 1975), represents the first of a new generation of satellites which have very great potential. These satellites have several important advantages: substantially higher resolution than the meteorological satellites, image format which is an accurate map projection, and multispectral (multiple wavelength) data which permit use of spectral pattern-recognition techniques. These advantages are gained, however, at the expense of frequent coverage; the Landsat satellite passes over a given spot on Earth only once in 18 days (or with two such satellites, every nine days).

The results of measuring snow extent with Landsat images have been very encouraging (Meier, 1975). The accurate map projection of Landsat images permits quick and easy comparison with drainage basin or topographic maps at scales of

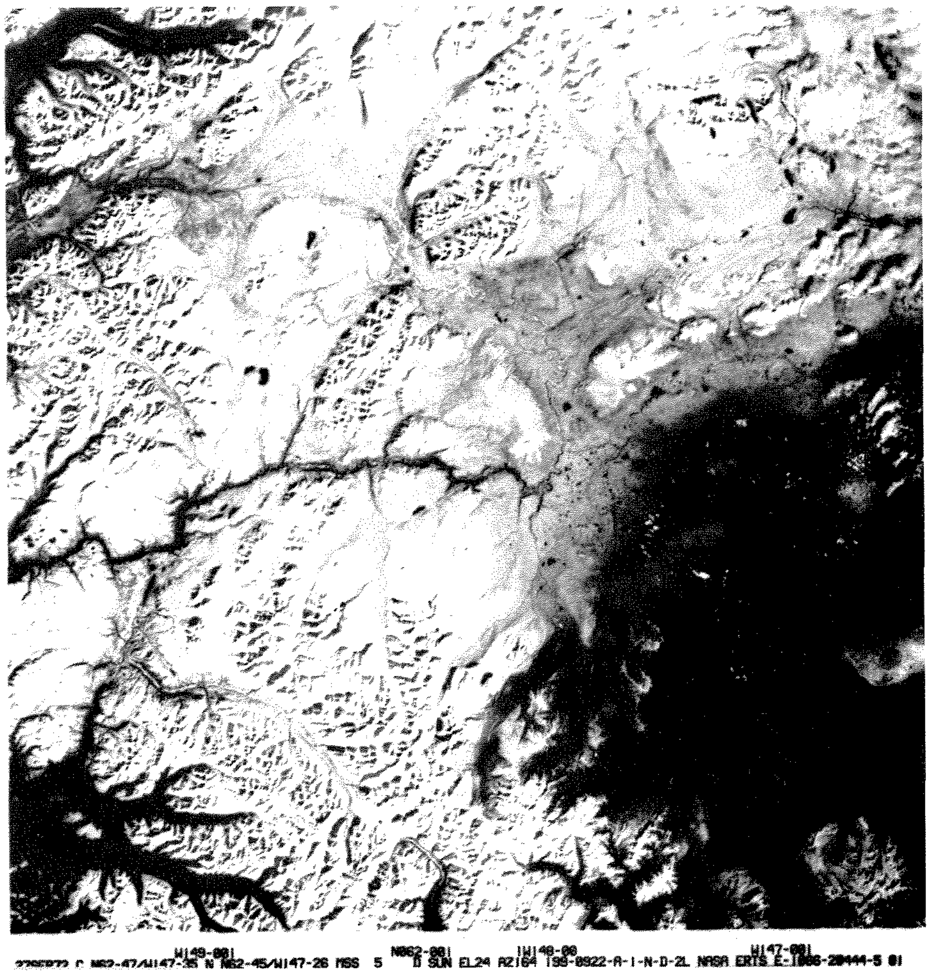

FIG. 2. Seasonal snow cover in a relatively non-mountainous environment. This Landsat view of the Gulkana basin, Alaska was taken on 27 September 1972. The tundra and taiga-covered plains and hills are covered with new snow except on the lower right cover. In the lower left corner a sharp snow line along mountain valleys is seen. In the right central area, some lakes are ice-free (black) while others are ice-covered (white). North is to the top of the page and the distance across this image is about 180 km. Landsat image ERTS E-1066-20444-5.

1:250 000 or 1:1 000 000. The resolution of the images is such that the changes in snow-covered area in a drainage basin as small as 6 km^2 can be measured easily (Krimmel & Meier, 1975). This is equivalent, for mapping purposes, of a resolution somewhat better than 100 m.

Satellite data in the visible, near infrared, and thermal infrared wavelengths, such as those obtained by either the NOAA or Landsat series of spacecraft, cannot be used routinely with computerized, automated procedures for measuring the areal extent of snow, because clouds and vegetation may obscure or mimic the signal from the snow. The radiant energy from snow and clouds often saturates visible or near infrared sensors, the multispectral signatures in the visible and near infrared wave-

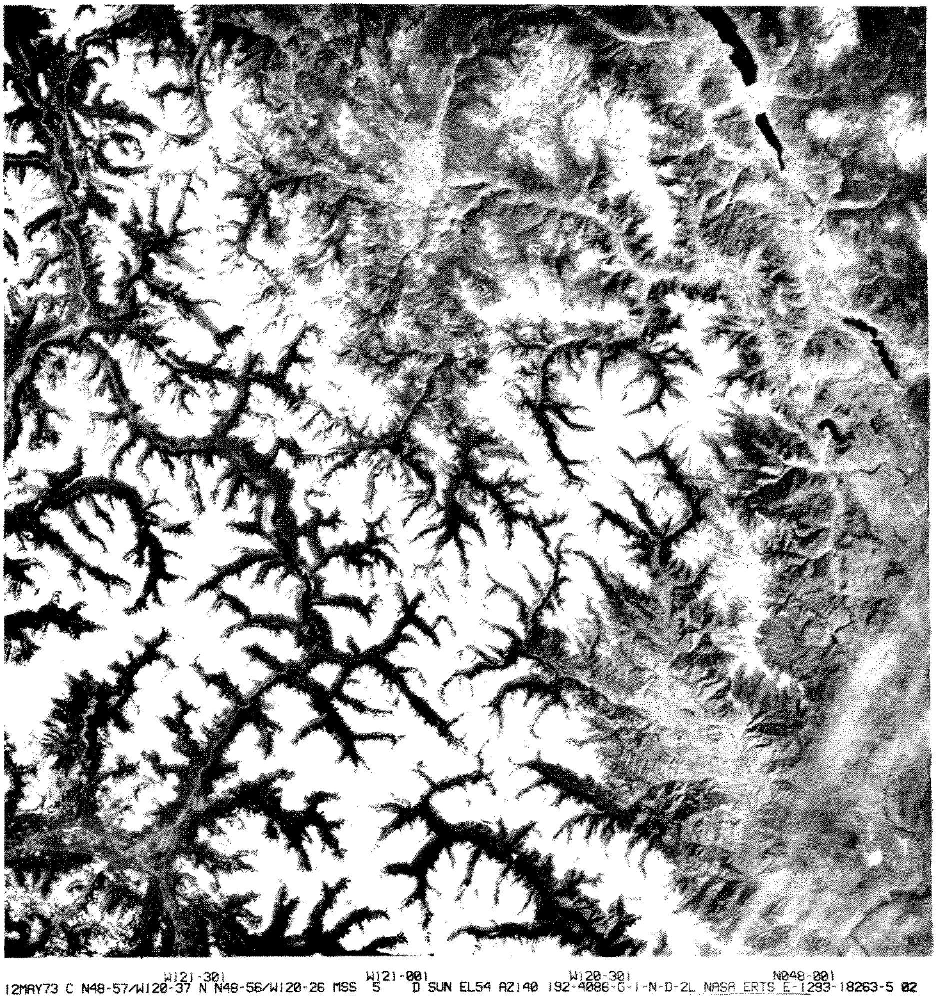

FIG. 3. Seasonal snow cover in a forested, mountain environment. This Landsat image is of the North Cascades in British Columbia, Canada (top) and Washington, USA (bottom). This is an area of extensive hydroelectric power and irrigation water supply development. Some snow exists under a heavy forest canopy and does not show in this satellite view. North is to the top of the page and the distance across the image is about 180 km. Landsat image ERTS E-1293-18263-5.

lengths of snow and clouds are similar, and clouds do not always have distinctive, machine-recognizable shapes. However, melting snow may appear appreciably darker than some clouds in certain near infrared wavelengths (Barnes *et al.*, 1974b). Much more information needs to be obtained about the multispectral reflectances of different types of snow and clouds before these results can be generalized. Clouds may also be distinguished from snow because the cloud-covered areas vary with time, so 'minimum brightness composites' over a period of several days may provide images only of the relatively unvarying snow cover (McClain & Baker, 1969).

Monitoring the areal extent of snow is impeded if the snow on the ground is masked by a vegetation canopy. If the canopy cover is not complete, then some radiant energy from snow may be detected by the satellite. In this case a multispectral analysis may be used to detect areas which combine the signatures of snow and of vegetation. However, if the vegetation canopy is completely closed, as in a dense conifer forest, the snow on the ground cannot be detected with visible, near infrared, or thermal infrared wavelength sensors.

Perhaps a solution to the problem of observing the areal extent of snow through clouds lies in the use of microwave radiation. Thermal radiation from snow at microwave wavelengths (passive microwave) can be detected by sensors carried in aircraft or in satellites. Clouds attenuate this radiation only slightly if relatively long wavelengths are used. The emission from snow is different from that of water, bare soil, or rock. Thus, the areal extent of snow cover should be measurable in any kind of weather by using microwave radiometers. One problem here is that the resolution of passive microwave radiometers will always be very coarse compared to reflected light sensors such as those used on Landsat. However, this may not be as severe a limitation as it appears; the microwave signal from snow, expressed as a brightness temperature, is so distinctive that the percentage of snow-covered area in any field of view can be calculated from the average brightness temperature observed (Meier, 1973b). So far, active microwave (radar) has not been useful for measuring the areal extent of snow (Meier, 1975).

Snow properties other than areal extent need to be measured for many scientific and practical purposes (UNESCO/IAHS/WMO, 1970). Most important is the snow mass per unit area, usually expressed as the water equivalent, and given by the product of average density and thickness. The snow mass per unit area can be measured directly by using gamma rays. The technique involves the measurement of natural gamma radiation by flying large scintillometers at low attitudes (50–150 m) above the ground. Most of the radiation detected at these altitudes originates from natural radioactivity in the bedrock or soil, and is attenuated by the snow cover. The attenuation is proportional to the snow mass. Thus the mass or water equivalent of the snow cover can be measured and averaged over the length of a flight line (Kogan *et al.*, 1965; Peck *et al.*, 1971; Grasty *et al.*, 1974). However, problems still exist with the operational use of this important tool: continuous monitoring requires continuous low-altitude flying and thus this method is limited to measurement of snow over the relatively flat terrain.

Monitoring of snowpack mass, especially over the mountainous areas, will require a different approach. Microwave radiation penetrates through snowpacks, and the emission/absorption caused by the snow is related, among other things, to the snow density and length of penetration through the snow thickness. Thus the potential for use of active or passive microwave radiation to measure snow mass is clear, but no system has yet been demonstrated that can do this.

Simple radar systems can be used to measure snow thickness or density (Vickers, 1971). Use of a single frequency, however, does not allow determination of both thickness and density, but if one parameter is known the other can be determined. A multi-frequency technique (Linlor, 1975) does show the possibility of determining both thickness and density (and therefore the water equivalent) of snow, and perhaps even the characteristics of a second underlying layer. However, such a system exists only in concept at this stage. It probably cannot be used from satellite altitudes, and the possible effects of scattering, surface roughness, gradational interfaces, etc. have yet to be evaluated.

Passive microwave techniques have been studied for many years as a method of determining snow properties from aircraft or satellites (Edgerton *et al.*, 1971; Meier & Edgerton, 1971). However, the resolution of these systems is limited by the physical size of antennae. For very large snow fields, such as the accumulation area of the Greenland ice sheet, this is not a problem, but microwave surveys for hydrological applications in small drainage basins may have to be conducted by aircraft in order to provide sufficient resolution.

The physics of microwave emission from snow is still not completely understood. Carefully controlled field experiments have shown that the microwave emission changes in a regular way as the snow pack thickness increases from zero to about a metre or so (Meier & Edgerton, 1971); thus it appears in principle that one could determine the water equivalent of snow from the microwave brightness. However, scattering of radiation within the snow volume strongly influences the radiation detected (England, 1974; Chang *et al.*, 1976; Ellerbruch *et al.*, 1977; Zwally, 1977). Unfortunately, the complex effect of scattering on microwave emission from snow is not yet well defined. The possibility of remotely measuring snow mass and other properties is so important that the problem of volume scattering deserves much further study.

The properties of temperature and wetness of snow are of primary interest to heat and water balance studies. Thermal infrared (wavelengths 3–5.5 and 8–14 μm) sensors yield information on the surface temperature and the surface melting condition. Also, combined visible and near infrared measurements may be used to determine whether or not the surface snow is melting. Unfortunately, thermal infrared (or combined visible/near infrared/thermal infrared) radiometers can measure effects only in the surface and near-surface layers of the snow. Passive microwave radiometers, on the other hand, respond to properties through several tens of centimetres of snow. The microwave emission from wet snow is markedly different from that of dry snow, but our present lack of understanding of volume scattering of microwave emission from snow prevents the use of this potential tool except in a very crude and qualitative way.

Glaciers and ice sheets

Glacier inventories require detection of snow and ice at a time when the seasonal snow cover is a minimum (UNESCO/IAHS, 1970a) (Fig. 4). Except in the required timing, the problem is similar to the measurement of the areal extent of snow. As most of the unknown or unmeasured glaciers are small, high resolution is required, and thus satellites of the Landsat type are required. Such satellites can be used to identify and classify glaciers as small as 100 m × 200 m (Higuchi, 1975). Satellite measurements may be very useful in correcting glacier outlines on old or inaccurate maps (R. S. Williams *et al.*, 1975; Williams, 1976; MacDonald, 1976).

Glacier mass balances can be approximated by measurement of the accumulation–area ratio (AAR) or equilibrium line altitude (ELA) (Meier & Post, 1962; UNESCO/IAHS, 1970b, 1973). Satellite data have been successfully used to measure AAR (Østrem, 1975); the problem is essentially the same as measuring the areal extent of seasonal snow cover. The passive microwave emission from polar ice sheets depends on the rate of snow accumulation and snow temperature, among other factors. The microwave emission and snow temperature can be measured by satellite sensors, so it appears to be possible to estimate the accumulation rate

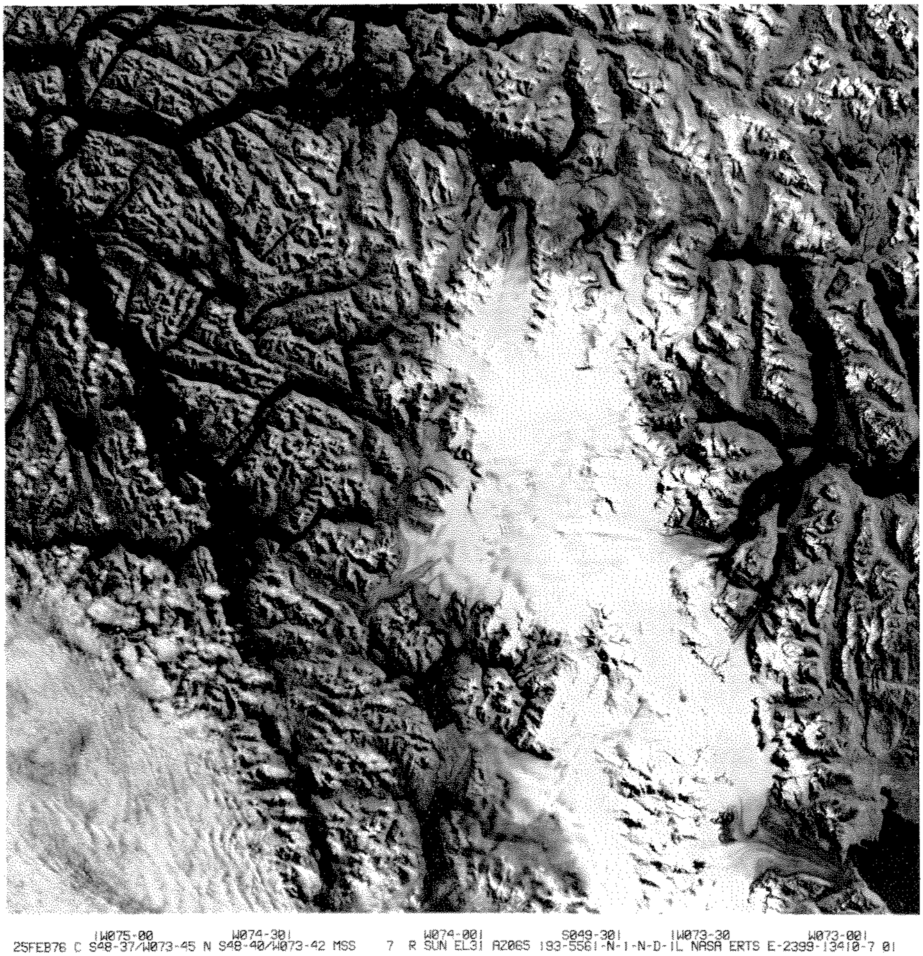

FIG. 4. *Glaciers of Hielo Patagonico Sur,* (the 'Southern Icefield') *of Patagonia, images such as this are* useful for making a glacier inventory, measuring glacier variations, and determining the accumulation–area ratio (AAR) of glaciers, as aerial photography of this area is difficult and expensive to obtain because of frequent poor weather. The calving terminus of the O'Higgins Glacier (right centre) retreated approximately 5 km between February 1974 and February 1976 as determined by Landsat images. The prominent boundaries between snow and firn, or firn layers of different ages, are enhanced by volcanic ash layers. North is to the top of the page and the distance across the image is about 180 km. Landsat image ERTS E-2399-13410-7, 25 February 1976.

(Zwally, 1977). Difficulties arise for those glaciers where appreciable meltwater is refrozen within the mass as internal accumulation or as superimposed ice. The internal accumulation probably cannot be measured using remote sensing, but it does appear to be possible to delineate the glacier ice/superimposed ice boundary. This possibility needs to be studied more carefully.

Glacier variations (UNESCO/IAHS, 1969a) may be monitored by satellite remote sensing if the variations exceed satellite resolution. Variations of some calving glaciers (Meier, 1973a), surging glaciers (Krimmel & Meier, 1975), and small ice caps (Williams *et al.*, 1974; Williams, 1976) (Fig. 5) have been successfully

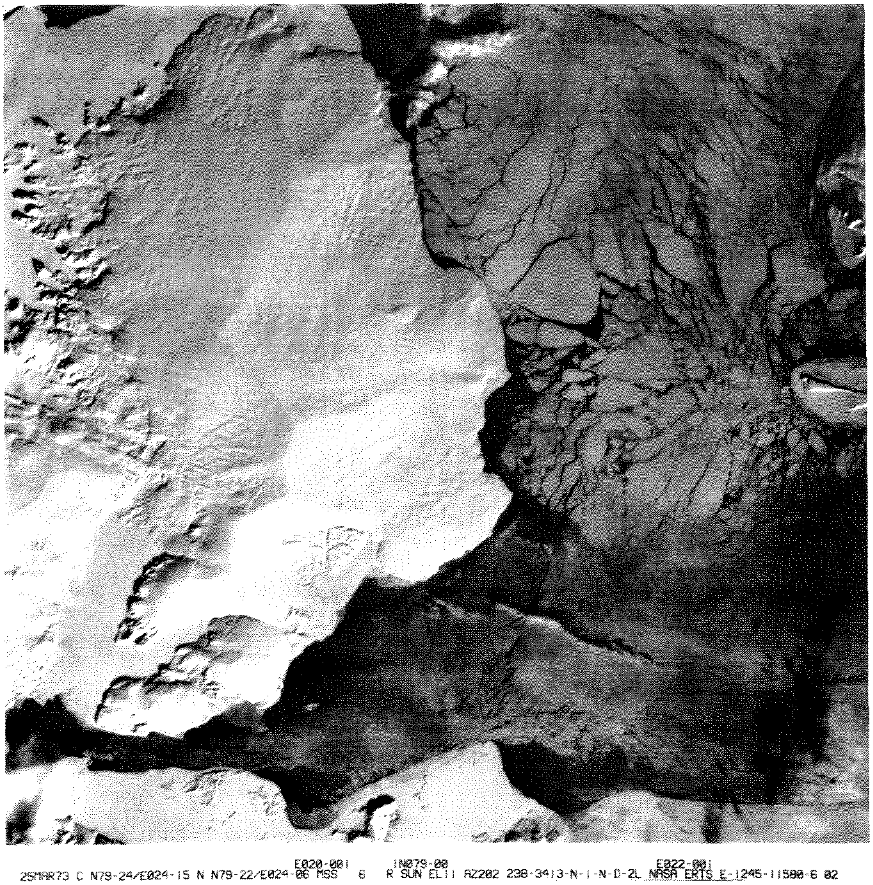

FIG. 5. Land ice and sea ice, Nordaustland, Svalbard (Spitzbergen). In the centre of the image is the eastern ice cap Austfonna. Low sun angle illumination (11° above the horizon) brings out subtle topographic detail on this ice cap. Fast ice occurs in Hinlopen Strait along the bottom of the image and in several fiords to the left and above (north of) Austfonna. Sea ice floes, some very large indicating recent breakup, occupy much of the right side of this image. Distance across this image is about 180 km. Landsat image ERTS E-1245-11580-6, 25 March 1973.

measured using satellite imagery. However, most glacier variations are so slow that the change in one year's time cannot be seen on satellite images. Similarly, glacier motion can in some instances be measured (Williams *et al.*, 1974; Krimmel & Meier, 1975; Post *et al.*, 1976). However, the motion must be rapid in order to exceed the inherent positioning error of about 100 m for ERTS (Landsat) type satellites, or about 1000 m for meteorological satellites.

Determining the thickness of a glacier or ice sheet is important for many types of programmes (UNESCO/IAHS, 1969b). This can now be done using remote sensing technology. For polar ice sheets (no liquid water present) this can be done from aircraft using conventional radar designs (Robin *et al.*, 1969; Gudmandsen, 1970; Bogorodskiy, 1975). Internal structure (Gudmandsen, 1975), ice temperature (Bogorodskiy *et al.*, 1970), and the character of the ice/rock interface (Oswald, 1975) have also been studied by radio-echo-sounding techniques. The ice velocity can be measured by mapping the characteristics of the radio-echo (especially the way the echo dies away in time), on two successive radar measurements from the moving glacier surface, because the shape of the die-away curve depends on the roughness character of the unmoving bedrock (Nye *et al.*, 1972). For temperate glaciers (those with liquid water present) the problem is more difficult because liquid water pockets scatter the radiation. This problem has been solved, however, using a very low frequency radar (Watts *et al.*, 1975; Watts & England, 1976; Björnsson, 1978). Glacier or ice sheet thickness measurements from satellite altitudes do not appear to be likely in the near future, but there is no theoretical reason why such a measurement system could not be designed.

Ground ice

Ice in the ground is difficult to observe by remote sensing methods. Thermal infrared sensors will detect freezing temperature at the ground surface, an observation which may be of importance, but the radiation observed by these sensors does not penetrate the ground. The electrical properties of frozen soil and unfrozen, damp soil are very different, so the depth to a frozen layer (such as permafrost) or the thickness of a surface frozen layer (seasonal frost) should in principle be detectable using an active or passive microwave system. However, the heterogeneous nature of soil, including the possible existence of liquid water in a mass of frozen soil, makes this problem almost intractable. One technique which has shown promising results involves analysis of the attenuation of a radio wave propagating along the ground (Blomquist, 1975). Specialized, experimental radars have also been used (e.g. Bertram *et al.*, 1972; Annan & Davis, 1976), but with mixed results.

Lake and river ice

Monitoring the areal extent of floating, fresh water ice (UNESCO/IAHS, 1972) is a problem similar in nature to the monitoring of snow cover. Water and bare ground both present considerable tonal or spectral contrast to either ice or snow as viewed by visible or near infrared sensors. One important problem, however, is that rivers and most lakes have small dimensions in comparison to the spatial resolution characteristics of many satellite sensors, so that monitoring requires high resolution sensors, such as those operated on Landsat, or by special processing of the images of meteorological satellites to emphasize fine detail (McGinnis & Schneider, 1978).

Sideways-looking airborne radar (SLAR or SLR) is very useful in identifying the extent and type of lake ice. High resolution along the sides of the flightline is obtained by using optical or computer processing to make a small radar antenna simulate the performance of a dimensionally much larger antenna. Such a 'synthetic aperture' radar system has been carried to satellite altitudes aboard Seasat-1 (launched 26 June 1978), which was able to acquire high-resolution (25 m) radar images from space. Radar images can be used to classify different types of lake ice, although young, smooth ice appears dark and indistinguishable from open water on some images (Page & Ramseier, 1975). Ice thickness can also be measured directly by a special type of aircraft radar designed for cm-length resolution (Page & Ramseier, 1975).

Ice concentration (the fraction of water surface covered by ice) could, in principle, be sensed directly from satellite altitudes by measuring where the passive microwave brightness temperature falls between that of continuous ice and open water. This promising method has yet to be properly evaluated.

In polar and subpolar regions, water trapped between a surface frozen layer and perennially frozen ground may continuously erupt to the surface in winter, forming a large ice accumulation called a naled (plural: naledi). Naledi (also called aufeis or

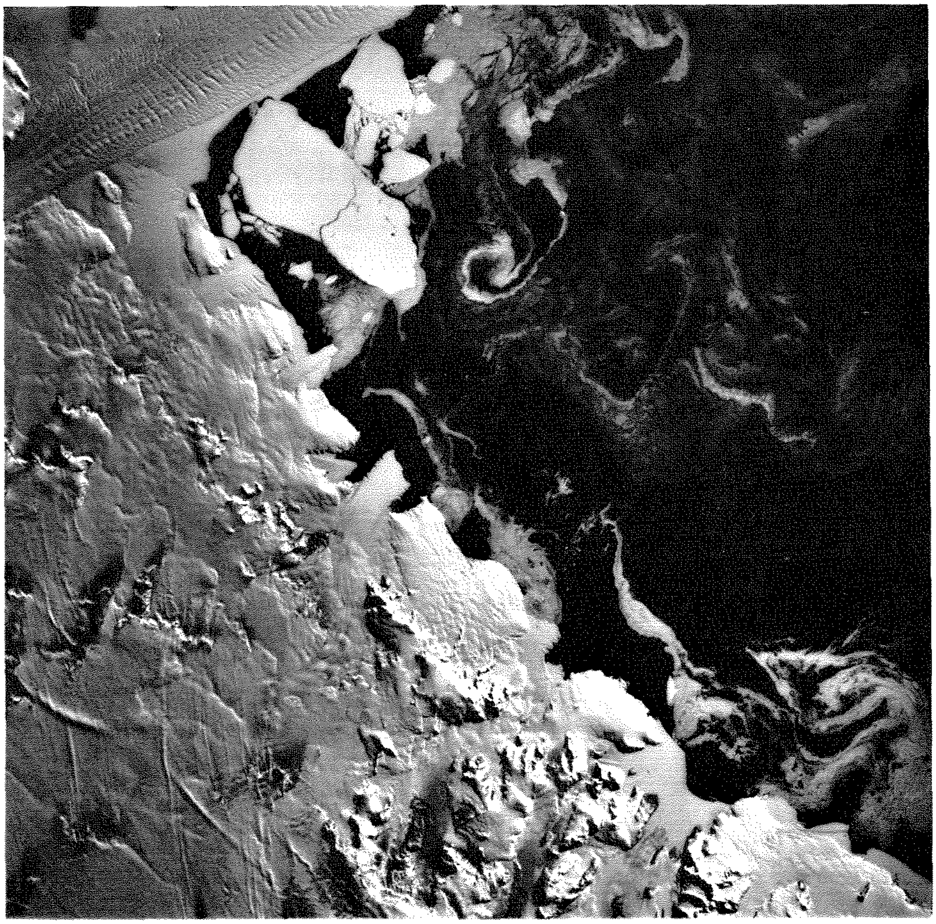

FIG. 6. A wide variety of ice types is shown on this Landsat (ERTS) view of part of the coast of Victoria Land, Antarctica. Isolated mountains (nunataks) break through the inland ice sheet in the lower left; over these appear streaks in two directions due to cirrus clouds. In the upper left corner, the huge Drygalski Ice Tongue flows out into the Ross Sea (right). The smaller Nordenskjöld Ice Tongue appears in the centre of this image. Published topographic maps, which were based on 1957–1961 aerial photography, show this ice tongue almost 22 km longer than it appears on this 22 February 1973 image, indicating a huge break-off. The Nordenskjöld Ice Tongue is the seaward continuation of the Mawson Glacier, which is typical of hundreds of small outlet glaciers along the margin of the Antarctic ice sheet; note the prominent flow ridges on this glacier. Fast ice (sea ice which freezes and remains in place along a shoreline) appears in Granite Harbour (lower right). Just below the Drygalski Ice tongue, a large mass of fast ice has broken loose and is beginning to crack up and float away. Sea ice floes in the Ross Sea display an interesting pattern of swirls and curliques (right). Distance across this image is about 180 km. Landsat image ERTS E-1214-20064-7, 22 February 1973.

icings) are important indicators of winter water supply and have pronounced environmental effects. They can be mapped using visible or near infrared sensors, because the naled surface is lighter in tone than the surrounding terrain in summer and darker in winter (Harden *et al.*, 1977). Also, the surface of a naled is warmer than the surrounding terrain in winter and thus can be detected by thermal infrared sensors.

Sea ice

Although sea ice is often not considered in hydrological studies, it is a very important element in the Earth's climate machine, and monitoring of sea ice is necessary for practical reasons in many areas (UNESCO/IAHS, 1972). Sea ice, like snow cover, changes rapidly in areal extent over time, and remote sensing methods are mandatory for accurate monitoring programmes (Fig. 6).

Visible, near infrared, and thermal infrared wavelengths are useful for mapping sea ice extent, observing surface melting, and following the motion or deformation of the ice cover (Barnes & Bowley, 1974; Campbell *et al.*, 1974; McClain, 1974; Campbell *et al.*, 1975a, b). Unfortunately, synoptic monitoring of sea ice using these wavelengths is hampered by persistent stratus clouds (which affect all wavelengths) in summer, and darkness (which affects all wavelengths except thermal infrared) in the winter.

Because of this limitation, microwave sensors are especially important for observing sea ice. Passive microwave radiometers can be used to distinguish ice from open water and multi-year ice from first-year ice (Gloersen *et al.*, 1973; Campbell *et al.*, 1974; Campbell *et al.*, 1975a; Tooma *et al.*, 1975). The electronically scanning microwave radiometer (ESMR) on the Nimbus-5 satellite now continually produces remarkable images showing a variety of ice types on land and sea in the polar regions (Fig. 7) (Zwally & Gloersen, 1977).

Radar, especially SLAR, is also very useful in determining characteristics of sea ice (Page & Ramseier, 1975; Dunbar, 1975). Coastal radar installations can be used to produce time-lapse movies of sea ice growth and movement (Tabata, 1970, 1975). A problem which remains, however, is the fact that smooth new ice produces very little radar return (a poor radar reflector) and therefore looks very much like open water.

Sea ice thickness is a very important characteristic to measure, because it is critical to numerical modelling of sea ice growth and deformation. Thickness can be inferred from ice-type identifications made with passive microwave, or from surface temperatures observed by thermal infrared sensors. Direct measurement is more difficult, although some special radars show promise (Page & Ramseier, 1975). Alternatively, the thickness can be estimated by measuring the amount of ice above water level with a laser altimeter (Hibler, 1975), or the amount of ice below water level with an upwards-directed sonar beam from a submarine (E. Williams *et al.*, 1975).

METHODS FOR ANALYSING REMOTE SENSING DATA

With the introduction of any major new technology such as remote sensing, new concomitant problems usually appear. Satellite remote sensing produces a veritable flood of raw data, the archiving and preliminary processing of which requires as

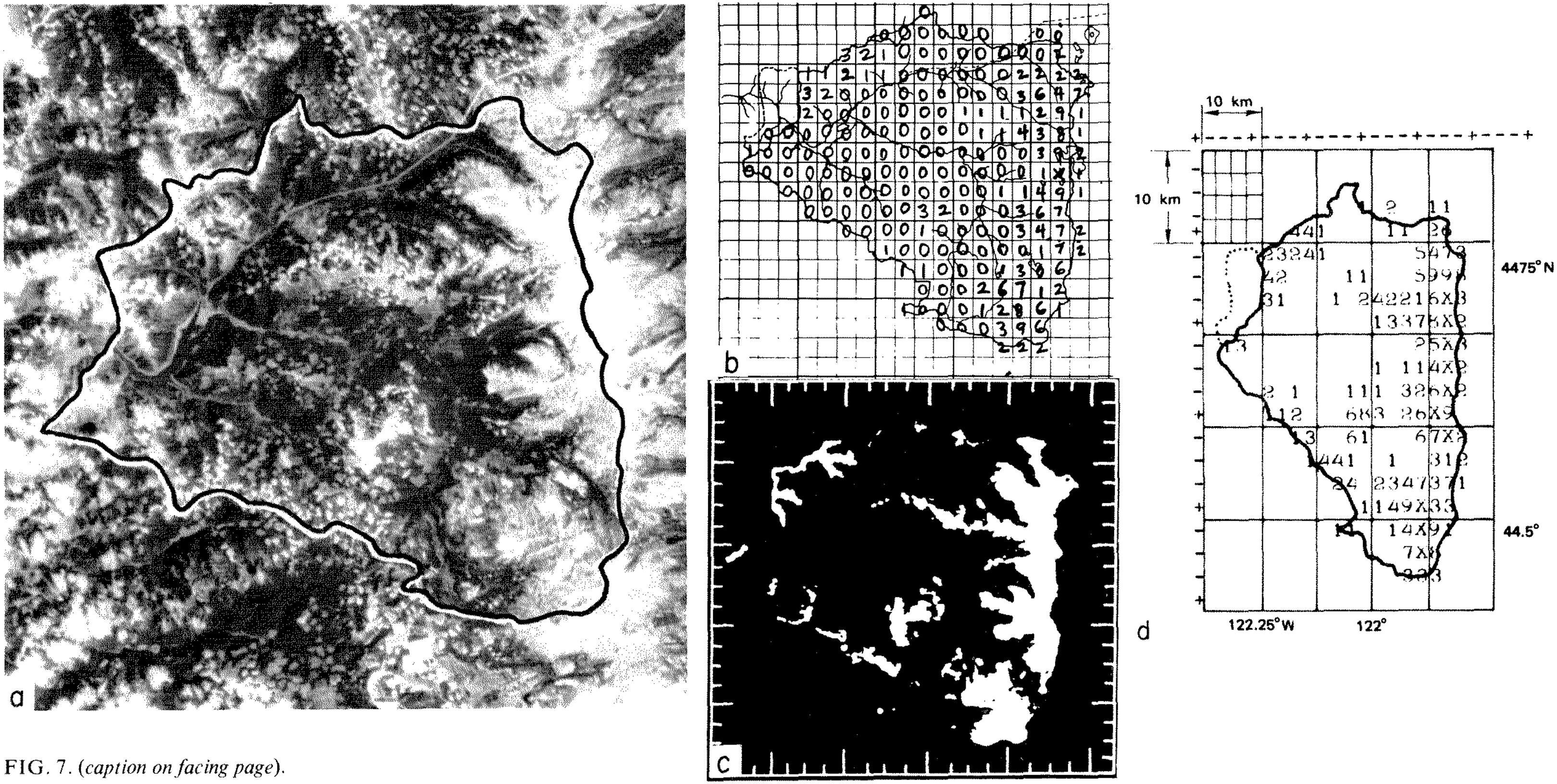

FIG. 7. (*caption on facing page*).

much (or sometimes far more) planning and study as does the actual data analysis leading to the experimental result. For instance, a single Landsat scene (4 or 5 bands), covering 186 km by 181 km on Earth, requires at least one reel of magnetic tape if it is to be stored in digital form.

Extracting quantitative information (such as areal coverage of a particular snow or ice form) from an image is probably the most important task, and is common to many studies, such as measuring the total hemispheric snow/sea ice cover for climatic analysis, measuring the snow area within a drainage basin for runoff forecasting, measuring AAR for glacier mass balance studies, observing the buildup and decay of naledi for water supply assessment, or measuring the amount of each type of sea ice in a shipping lane. The various methodologies useful to these studies are discussed further in Meier & Evans (1975).

The simplest way to measure area on a satellite image is simply to draw a line around the object of interest, and then measure the circumscribed area with a planimeter or by counting dots (Barnes & Bowley, 1968). Many satellite images, however, do not represent a true map projection; hence, this simple method fails. Optical devices, such as a zoom transfer scope, can be used to stretch one dimension only, thereby partly restoring the image to a geometrically correct map base. Mapping of snow cover in major river basins is being done rapidly and routinely with a zoom transfer scope and an electronic planimeter for rapid area determination (Schneider, 1975). This method is relatively slow if the snow or ice boundary is very convoluted or patchy. It also offers little opportunity to compare the satellite image with data from other sources to help resolve problem areas. The snowline in a mountainous area is often strongly related to altitude. Thus the image can be compared to a topographic map by means of a transparent overlay and the altitude of the snowline read off directly. This is a very rapid method, and has an added advantage in that if the snowline is obscured locally by clouds, the snowline can be interpolated by following the altitude contour.

An extremely useful method is to grid the image (Fig. 7). If an overlay is prepared with grid cells of an appropriate size, the snow or ice cover within each cell can be estimated by eye very quickly. For instance, working with 1:250 000 enlargements of Landsat images, a cell size of 1 cm^2 (2.5 km $\times$ 2.5 km) is convenient (Meier & Evans, 1975). Also this method makes comparison of several images (such as snow-covered and snow-free) convenient.

The contrast on an image between snow or ice and snow-free land or water may be so large that the snow or ice area can be separated by 'slicing' the image at a

FIG. 7. Examples of techniques used to extract snow-covered area data for hydrological purposes (from Meier & Evans, 1975). (a) Landsat image showing the drainage basin (outlined) of the North Santiam River above Detroit Reservoir, Oregon. The drainage basin area is 1134 km^2 in area and it is about 42.5 km wide and high. Landsat image ERTS-E-1257-18280-5, 6 April 1973. (b) Example of a manual technique for measuring snow area. The snow cover in 2.5 $\times$ 2.5 km grid boxes has been estimated in tenths (X = 10/10, 9 = 9/10, etc). (c) Example of a snow area mask created by density slicing, followed by visual editing. This was done on the Stanford Research Institute's electronic satellite image analysis console (ESIAC). (d) Snow-area coverage in tenths in 2.5 $\times$ 2.5 boxes has been recorded in computer files from the mask in (c), and thus these data can be efficiently related to altitude, other images, and so forth. This illustration was taken directly from a computer printout with unequal vertical and horizontal spacing, so the drainage basin appears to be compressed horizontally.

certain critical density (radiance). Density-slicing can be accomplished by using isodensity photographic film or a scanning densitometer. Alternatively, an electronic console can be used that views an image with a television camera or accepts digital input, and projects the image on a television screen where the contrast and other image properties can be manipulated or edited. The density-slicing method must be used with great caution unless editing can be done with reference to other sources of information. Usually there is an indistinct (fuzzy) snow/no-snow boundary. The exact setting within this transition zone cannot be determined easily, and a slight error in setting can produce very large errors in area. Thus it is important to bring as many additional data as possible into the decision about the appropriate threshold.

Electronic consoles can be used to combine density-slicing, time-lapse sequences, multi-spectral (colour) information, and other data to provide a quick yet very accurate determination of snow/ice area (Evans, 1973). Such consoles are not yet in common use, however.

Computer-assisted analysis from digital (computer compatible) tapes is usually the most precise, the most versatile, and the most expensive analysis method. Some satellite data are readily available in digital form. The first problem is locating the area of interest in a stream of digital information. This normally requires playing the data through a special image console. Alternatively, a crude image can be produced on standard computer printout pages, but may have to be so large to be readable that the method may not be practical. After identification and delineation of the area of interest, the ice or snow cover can often be identified by the computer device through comparison of the variation in the spectral radiance of specific phenomena at several different wavelengths. In the case of Landsat, 4 different wavelengths (spectral bands) can be used, which is usually sufficient for classifying snow and ice. However, the correct comparisons have to be 'taught' to the computer through the use of 'training sets'. This may require an interactive link, between the computer device, the output console, and the scientist.

PROBLEMS FOR FUTURE RESEARCH

Many questions need to be answered before the rapidly varying snow and ice resources of Earth can be adequately and accurately monitored by remote sensing techniques. A new generation of Earth resources, oceanographic, and meteorological satellites will soon be launched. These new types of satellites will provide more frequent coverage, more multi-spectral channels of different wavelength bands in the visible, near infrared, and thermal infrared wavelengths, and much higher resolution than existing civilian, scientific satellites. In order to exploit these exciting new possibilities fully, we must be able to handle the huge data and information flow which will result; thus, we must turn more and more to automatic (machine) processing of the imagery. The basic research necessary for this has not yet been accomplished; we can automatically discriminate snow from wheat fields, alluvial fans, or open water, but not from clouds. We are not even sure if such a discrimination is possible. Increased research in this field is obviously necessary for both short and long term benefits.

In the more distant future we will require, and probably be able to obtain, an all-weather capability for observing ice and snow at resolutions appropriate to many hydrological problems and to many types of hydrogeological and hydrometeorological phenomena. This will require active or passive microwave sensors superior in design to those which exist today. The physics of emission, reflection, and scattering from ice and snow packs is not yet well known, either in theory or by empirical observation. Many questions need to be answered before a proper active or passive microwave sensing system can be designed. These include such matters as the interface/volume scattering properties of snow (what *are* the objects in snow which cause volume scattering?). Also, what are the dielectric and scattering properties of wet snow, and salty ice, and what are the electrical properties of the thin film of water which occurs on ice grains? Thus, many fertile fields of research endeavour exist, the solution of which will eventually allow us to monitor accurately such an important element of our environment: the global distribution of snow and ice.

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